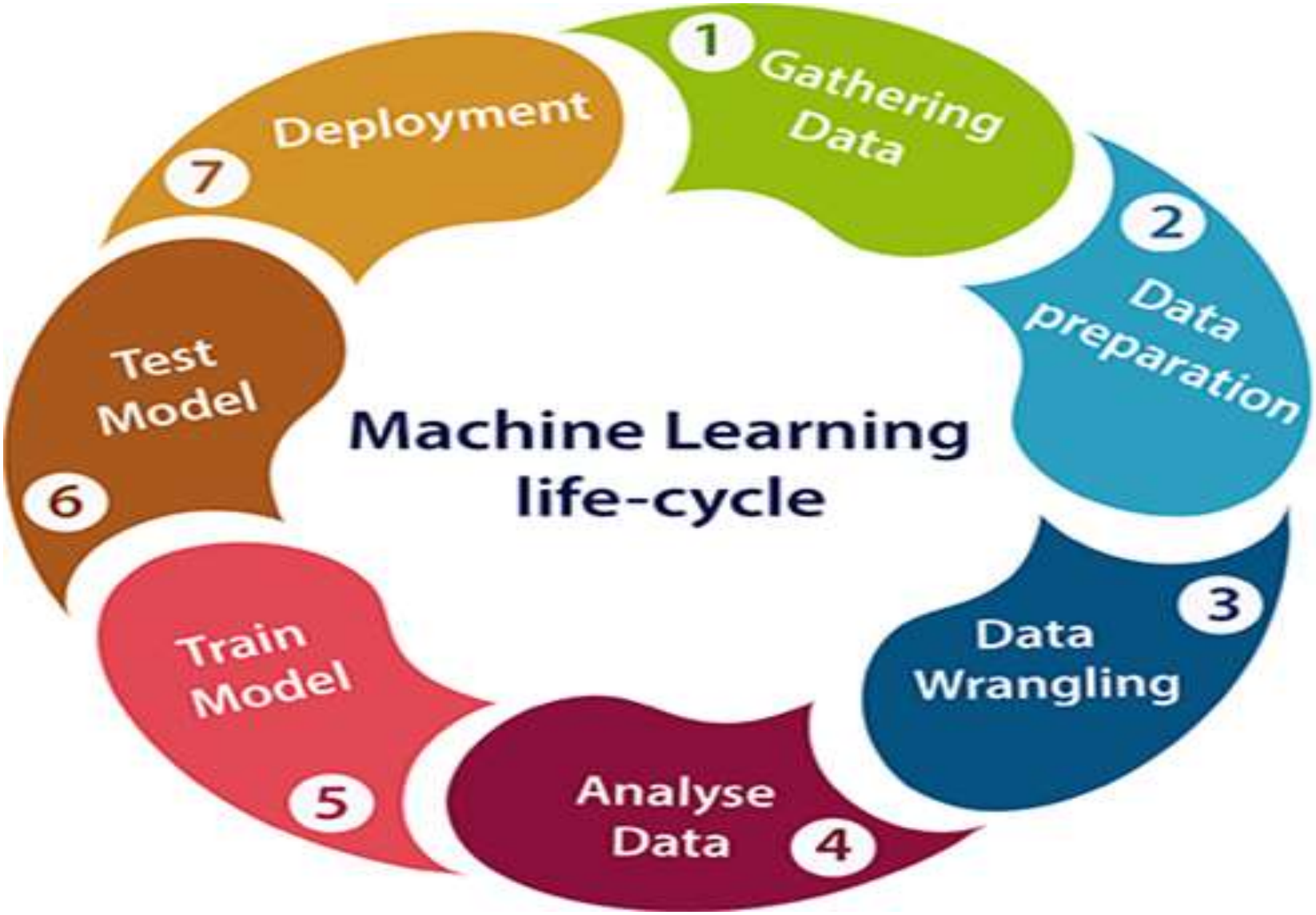
Unit-2
Preparing to Model

- Machine Learning activities,
- Basic Types of data in Machine Learning,
- Structures of data,
- Data Quality and Remediation,
- Data Pre-Processing,

Machine Learning Activities



Activities

- Gathering Data
- Data preparation
- Data Wrangling
- Analyse Data
- Train the model
- Test the model
- Deployment

1. Gathering Data

- The goal of this step is to identify and obtain all data-related problems.
- In this step, we need to identify the different data sources, as data can be collected from various sources such as **files**, **database**, **internet**, or **mobile devices**.
- It is one of the most important steps of the life cycle.
- The quantity and quality of the collected data will determine the efficiency of the output.
- The more will be the data, the more accurate will be the prediction.

- This step includes the below tasks:
  - **Identify various data sources**
  - **Collect data**
  - **Integrate the data obtained from different sources**
- By performing the above task, we get a coherent set of data, also called as a **dataset**. It will be used in further steps.

2. Data preparation

- After collecting the data, we need to prepare it for further steps.
- **Data preparation is a step where we put our data into a suitable place and prepare it to use in our machine learning training.**
- In this step, first, we put all data together, and then randomize the ordering of data.
- This step can be further divided into two processes:
- **Data exploration:**
  - It is used to understand the nature of data that we have to work with.
  - We need to understand the characteristics, format, and quality of data.
  - A better understanding of data leads to an effective outcome. In this, we find Correlations, general trends, and outliers.
- **Data pre-processing:**
- Now the next step is preprocessing of data for its analysis.

3. Data Wrangling

- Data wrangling is the **process of cleaning and converting raw data into a useable format.**
- It is the **process of cleaning the data, selecting the variable to use, and transforming the data in a proper format** to make it more suitable for analysis in the next step.
- It is one of the **most important steps of the complete process**.
- Cleaning of data is required to address the quality issues.
- It is not necessary that data we have collected is always of our use as some of the data may not be useful. In real-world applications, collected data may have various issues, including:
  - **Missing Values**
  - **Duplicate data**
  - **Invalid data**
  - **Noise**
- So, we use various **filtering techniques to clean the data.**
- It is **mandatory to detect and remove the above issues because it can negatively affect the quality of the outcome.**

4. Data Analysis

- Now the cleaned and prepared data is passed on to the analysis step. This step involves:
  - **Selection of analytical techniques**
  - **Building models**
  - **Review the result**
- The aim of **this step is to build a machine learning model to analyze the data using various analytical techniques and review the outcome**.
- It starts with the determination of the type of the problems, where we select the machine learning techniques such as **Classification**, **Regression**, **Cluster analysis**, **Association**, etc. then build the model using prepared data, and evaluate the model.
- Hence, in this step, **we take the data and use machine learning algorithms to build the model.**

5. Train Model

- Now the next step is to train the model, in **this step we train our model to improve its performance for better outcome of the problem.**

- We **use datasets to train the model using various machine learning algorithms**.

- Training a model is required so that **it can understand the various patterns, rules, and, features.**

6. Test Model

- Once our machine learning model has been **trained on a given dataset, then we test the model.**

- In this step, **we check for the accuracy of our model by providing a test dataset to it.**

- Testing the model **determines the percentage accuracy of the model as per the requirement of project or problem.**

7. Deployment

- The last step of machine learning life cycle is deployment, **where we deploy the model in the real-world system.**

- If the above-prepared model is producing an accurate result as per our requirement with acceptable speed, then we deploy the model in the real system.

- **But before deploying the project, we will check whether it is improving its performance using available data or not**.

- The deployment phase is similar to **making the final report for a project.**

Basic Data types in ML

- Data can broadly be divided into following two types.
  - **Qualitative data**
  - **Quantitative data**

- **<u>Qualitative data</u>**
- divided it provides information about the quality of an object or information which cannot be measured.
- For example, if we consider the quality of performance of students in terms of '**Good**', '**Average**', and '**Poor**', it falls under the category of qualitative data.
- Qualitative data is **also called categorical data**.
- Qualitative data divided into 2 parts,
  - **Nominal data**
  - **Ordinal data**

- **Nominal data**
- It is one which has **no numeric value**, but a named value.
- It is used for **assigning named values to attributes**.
- Nominal values **cannot be quantified**.
- For examples,
  - **Blood group**: A,B,O,AB, etc
  - **Nationality**: Indian, American, British, etc
  - **Gender**: Male, Female, Other

- We **can not do any mathematical operations** on nominal data such as mean, variance, etc.

Are you married?

- Yes
- No

What languages do you speak?

- Englisch
- French
- German
- Spanish

- **Ordinal data**
- It is used to possessing the properties of nominal data, can also be natural ordered.
- This means the ordinal data also **assigns named values to attributes** but unlike nominal data, they can be arranged in a sequence of increasing or decreasing value so that we can say **whether a value is better than or greater than another value.**
- **For examples,**
  - **Customer satisfaction:** 'Very Happy', 'Happy', 'Unhappy', etc
  - **Grades:** A,B,C,etc
- **Median and quartiles** can be identified but **mean can still not identified.**

What Is Your Educational Background?

- ○ 1 - Elementary
- ○ 2 - High School
- ○ 3 - Undegraduate
- ○ 4 - Graduate

Quantitative Data

- It is relates **to information about the quantity of an object**. Hence, **it can be measured.**

- **Quantitative Data (**Continuous Data) represents measurements and therefore their values **can't be counted but they can be measured**.

- **For example**, if we consider the **attributes 'marks'**, it can be measured.

- Quantitative data is also termed as numeric data.

- There are two types of Quantitative Data,
  - **Interval Data**
  - **Ratio Data**

- **Interval Data**
- Interval data is numeric data for which not only the order is known, but the exact difference b/w values is known.
- **For example**, Celsius Temperature.

- The difference b/w 12C and 18C degrees is 6C and same as for 15.5C and 21.5C.
- other examples include data, time, etc.

- For interval data, We **can do any mathematical operations** on interval data such as mean, median , mode, variance, SD etc.

- **Data don't have 'true zero' value**.
- **For example**, we can not say '0 temperature' or 'No temperature'.

Temperature?

- - 10
- -5
- 0
- + 5
- + 10
- + 15

- **Ratio Data**
- It represents numeric data for **which exact value can be measured**.
- **Absolute zero is available for ratio data**.
- On These variables we can do mathematical operations.
- **For example**, Height, weight, age, salary, etc

Length (inch)?

- ○ 0
- ◉ 5
- ○ 10
- ○ 15

S.N.	Character	Quantitative Data	Qualitative Data
1.	**Definition**	These are data that deal with quantities, values, or numbers.	These data, on the other hand, deals with quality.
2.	**Measurability**	Measurable.	They are generally not measurable.
3.	**Nature of Data**	Expressed in numerical form.	They are descriptive rather than numerical in nature.
4.	**Research Methodology**	Conclusive	investigative
5.	**Quantities measured**	Measures quantities such as length, size, amount, price, and even duration.	Narratives often make use of adjectives and other descriptive words to refer to data on appearance, color, texture, and other qualities.
6.	**Data Structure**	Structured	Unstructured

Data structures for machine learning

Auto MPG data set

1	mpg	cylinders	displacement	horsepower	weight	acceleration	model year	origin	car name
2	18	8	307	130	3504	12	70	1	chevrolet chevelle malibu
3	15	8	350	165	3693	11.5	70	1	buick skylark 320
4	18	8	318	150	3436	11	70	1	plymouth satellite
5	16	8	304	150	3433	12	70	1	amc rebel sst
6	17	8	302	140	3449	10.5	70	1	ford torino
7	15	8	429	198	4341	10	70	1	ford galaxie 500
8	14	8	454	220	4354	9	70	1	chevrolet impala
9	14	8	440	215	4312	8.5	70	1	plymouth fury iii
10	14	8	455	225	4425	10	70	1	pontiac catalina
11	15	8	390	190	3850	8.5	70	1	amc ambassador dpl
12	15	8	383	170	3563	10	70	1	dodge challenger se
13	14	8	340	160	3609	8	70	1	plymouth 'cuda 340
14	15	8	400	150	3761	9.5	70	1	chevrolet monte carlo
15	14	8	455	225	3086	10	70	1	buick estate wagon (sw)
16	24	4	113	95	2372	15	70	3	toyota corona mark ii
17	22	6	198	95	2833	15.5	70	1	plymouth duster
18	18	6	199	97	2774	15.5	70	1	amc hornet

Dataset Description

Column Position	Attribute Name	Description	Examples	Attribute Type	Nulls Ratio
#1	mpg	fuel efficiency measured in miles per gallon (mpg)	9.0, 13.0, 41.5	quantitative	0%
#2	cylinders	number of cylinders in the engine	3, 4, 8	qualitative	0%
#3	displacement	engine displacement (in cubic inches)	68.0, 112.0, 455.0	quantitative	0%
#4	horsepower	engine horsepower	46.0, 70.0, 230.0	quantitative	2%
#5	weight	vehicle weight (in pounds)	1613, 3615, 5140	quantitative	0%
#6	acceleration	time to accelerate from O to 60 mph (in seconds)	8.00, 15.50, 24.80	quantitative	0%
#7	model year	model year	73, 79, 82	qualitative	0%
#8	origin	origin of car (1: American, 2: European, 3: Japanese)	1, 2, 3	qualitative	0%
#9	car name	car name	audi fox, subaru	qualitative	0%

DATA REMEDIATION

- Data remediation is a **part of data quality.**
- Data remediation is **an activity that's focused on cleansing, organizing and migrating data so it's fit for purpose or use.**
- The **process typically involves detecting and correcting** (or removing) corrupt or inaccurate records by **replacing**, **modifying** or **deleting the "dirty" data.**
- It can be performed manually, **with cleansing tools, as a batch process (script), through data migration or a combination of these methods.**

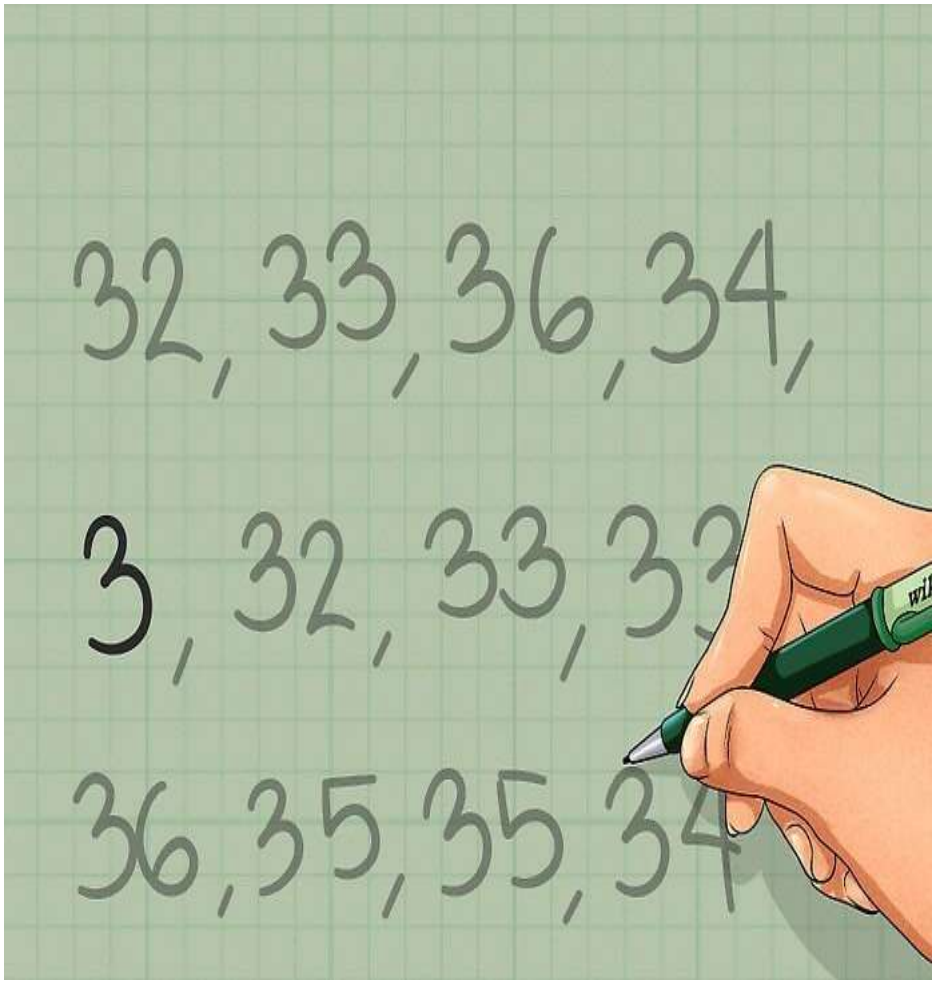
1. Handling outliers

- An outlier is a piece of data that is an abnormal distance from other points.

- In other words, it's data that lies **outside the other values** in the set.

- **Outliers can have many causes, such as**:
  - Measurement or input error.
  - Data corruption.

Example of outlier



Players	Scores
Player1	500
Player2	350
Player3	10
Player4	300
Player5	450

- **Remove outliers:** if the number of records which are outliers is not many, a simple approach may be to remove them.

- **Imputation :** one other way is to impute (assign) the value with mean or median or mode. The value of the most simmiler data element may also be used for imputation.

- **Capping :**For the values that lie outside the 1.5[x] IQR (interquartile range) limits, we can cap them by replacing those observations below the lower limit value of 5th percentile and those that lie above the upper limit, with the value of 95th percentile.

Data Pre-Processing

1. Dimensionality Reduction

- In machine learning classification problems, there are often too many factors on the basis of which the final classification is done.
- These factors are basically variables called features.
- **The higher the number of features, the harder it gets to visualize the training set and then work on it.**
- Sometimes**, most of these features are correlated, and hence redundant (unnecessary).**
- This is where dimensionality reduction algorithms come into play.
- Dimensionality reduction refers to reducing the number of input variables for a dataset**, by obtaining a set of principal variables.**
- It can **be divided into feature selection** and **feature extraction**.

- If your data is represented using rows and columns, such as in a spreadsheet, **then the input variables are the columns that are fed as input to a model to predict the target variable. Input variables are also called features.**

- We can consider the **columns of data representing dimensions on an *n*-dimensional feature space** and **the rows of data as points in that space.** This is a useful statistical analysis of a dataset.

- It is often desirable to reduce the number of input features. This reduces the number of dimensions of the feature space, **hence the name "*dimensionality reduction*."**

- An intuitive **example of dimensionality reduction** can be discussed through a simple e-mail classification problem, where we need to classify whether the e-mail is spam or not.

- This can involve a large number of features, such as whether or not the e-mail has a generic title, the content of the e-mail, whether the e-mail uses a template, etc.

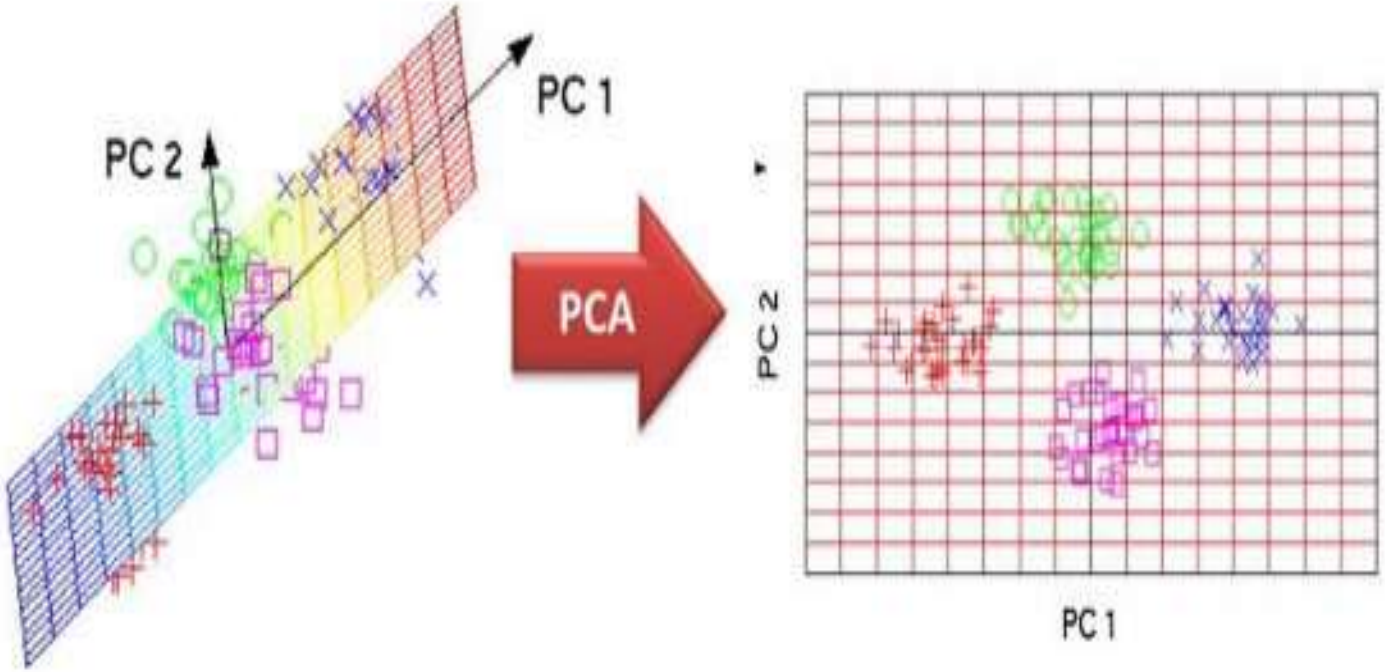
- The most **common approach to dimensionality reduction is called principal components analysis or PCA.**

- **It makes the large data set simpler, easy to explore and visualize.**

- Principal Component Analysis(PCA) is one of the most popular linear dimension reduction. Sometimes, it is used alone and sometimes as a starting solution for other dimension reduction methods.

- **PCA is a projection based method which transforms the data by projecting it onto a set of orthogonal (**right angles**) axes.**

Dimensionality Reduction
&
Principal Component Analysis



2. Feature subset selection

- Feature subset selection or simply called feature selection, both for supervised as well as unsupervised learning, try to find out the optimal subset of the entire feature set which significantly reduces computational cost without any major impact on the learning accuracy.

- **Feature Subset Selection** Methods can be classified into three broad categories…
  - **Filter Methods**
  - **Wrapper Methods**
  - **Embedded Methods**

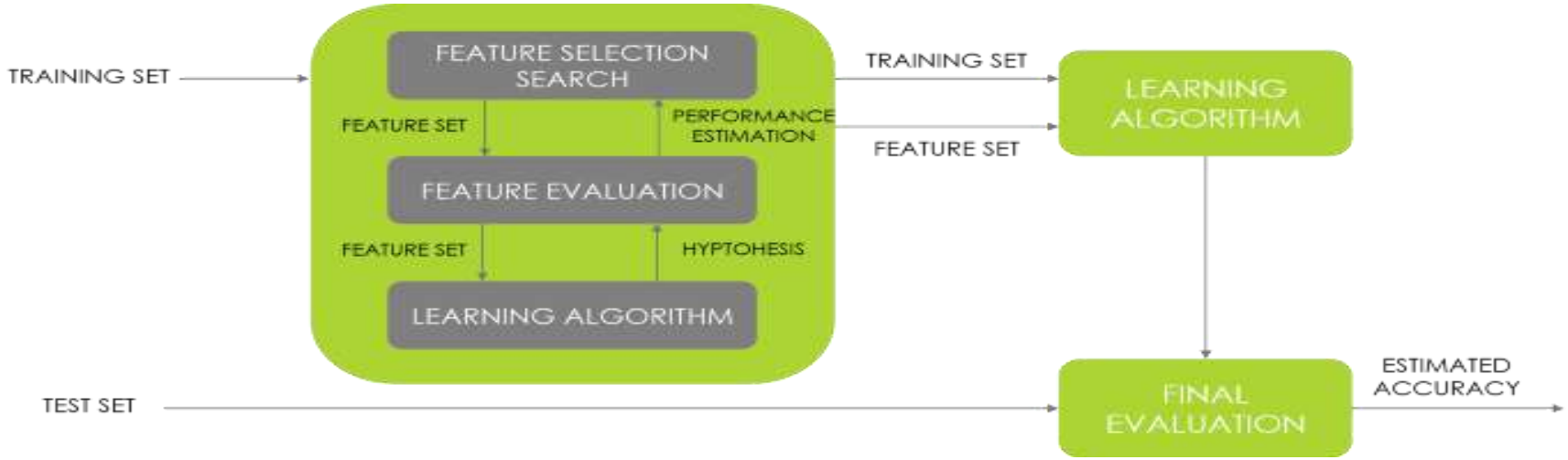
Filter Methods

- In this method, select subsets of variables as a pre-processing step, independently of the used classifier
- It would be worthwhile to note that **Variable Ranking**-***Feature Selection*** is a Filter Method.

- Filter Methods are usually fast.
- Filter Methods provide generic selection of features, not tuned by given learner (universal).

Wrapper Methods

- In Wrapper Methods, the **Learner is considered a black-box**.
- Interface of the black-box is used to score subsets of variables according to the predictive power of the learner when using the subsets.



- Results vary for different learners.
- One needs to define: – how to search the space of all possible variable subsets ?– how to assess the prediction performance of a learner ?

Embedded Methods

- Embedded Methods are specific to a given learning machine
- Performs variable selection (implicitly) in the process of training
- E.g. WINNOW-algorithm (linear unit with multiplicative updates).

Lazy vs. Eager Learning

- **Lazy learning** : Simply stores training data (or only minor processing) and waits until it is given a test tuple.

- Just store Data set **without** learning from it
- Start classifying data when it receive **Test data**
- So it takes less time learning and more time classifying data
- **e.g.** K - Nearest Neighbour, Case - Based Reasoning

- **Eager learning** : Given a set of training set, constructs a classification model before receiving new (e.g., test) data to classify.

- When it receive data set it starts classifying (learning)
- Then it does not wait for test data to learn.
- It's fast as it has pre-calculated algorithm.
- So it takes long time learning and less time classifying data.
- **e.g.** Decision Tree, Naive Bayes, Artificial Neural Networks